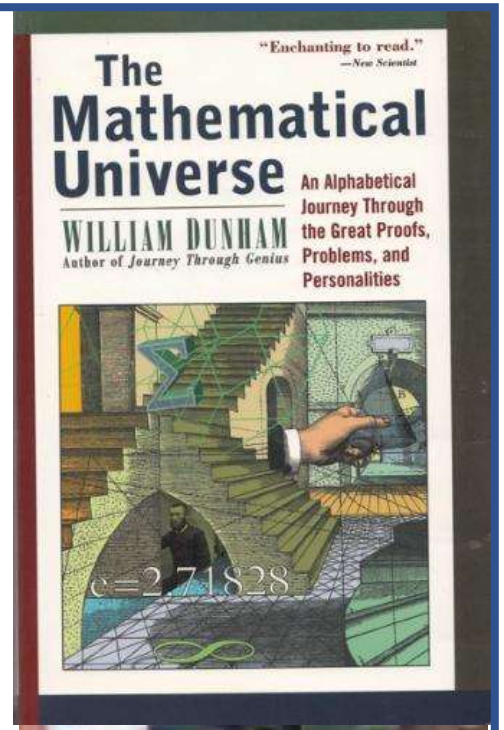
Course design of GFN1000

- Part 1. Human exploration of the physical universe
- Part 2. Human exploration of the world of life
- Part 3. Our understanding of human understanding
  - Text 7: Poincaré, *Science and Method*
  - Text 8: Kandel, *In Search of Memory*
  - Text 9: Needham, *The Shorter Science and Civilization in China*
  - Text 10a: Nathan Sivin, ‘Why the Scientific Revolution Did Not Take Place in China – or Didn’t it?’
  - Text 10b: 沈括:《梦溪笔谈》
  - Text 11a: Dunham, *The Mathematical Universe*
  - Text 11b: Euclid, *Elements*

GFN 1000
In Dialogue with Nature

Dunham & Euclid

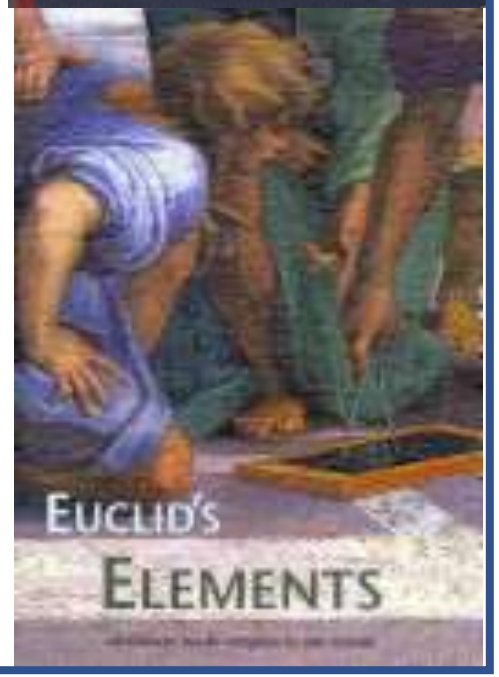
Texts and Core Question



Text 11a: William Dunham

The Mathematical Universe: *An Alphabetical Journey Through the Great Proofs, Problems, and Personalities* (Wiley 1994).

Ch. G - Greek Geometry



Text 11b: Euclid

"***Elements***"**:** **Book 1** (Green Lion Press, 2010)

What makes modern science so unique?

Origin of Western Mathematics

Where did geometry come from?

- GEO + METRY → Land + Measurement
- Greek geometry ranks as a major achievement of the human intellect for reasons both mathematical and historical, both practical and aesthetic.
- Greeks learned mathematics from the Egyptians and the Babylonians.
  - *"There is no question that* ***geometric ideas predated the Greeks****. The civilizations of Egypt and Mesopotamia, for instance, used geometry to partition fields and erect pyramids."* [Text 11a, para. 6]
- *"But it is to the Greeks that we look for the first geometric theorems, the first propositions proved with logical rigor."* [Text 11a, para. 6]

Pythagorean school (毕氏学派)

- Greek philosopher, mathematician, founder of the religious movement



Pythagoras
(c. 570 – c. 495 BC)

The School of Athens (1511) - Raphael





Pythagorean school (毕氏学派)

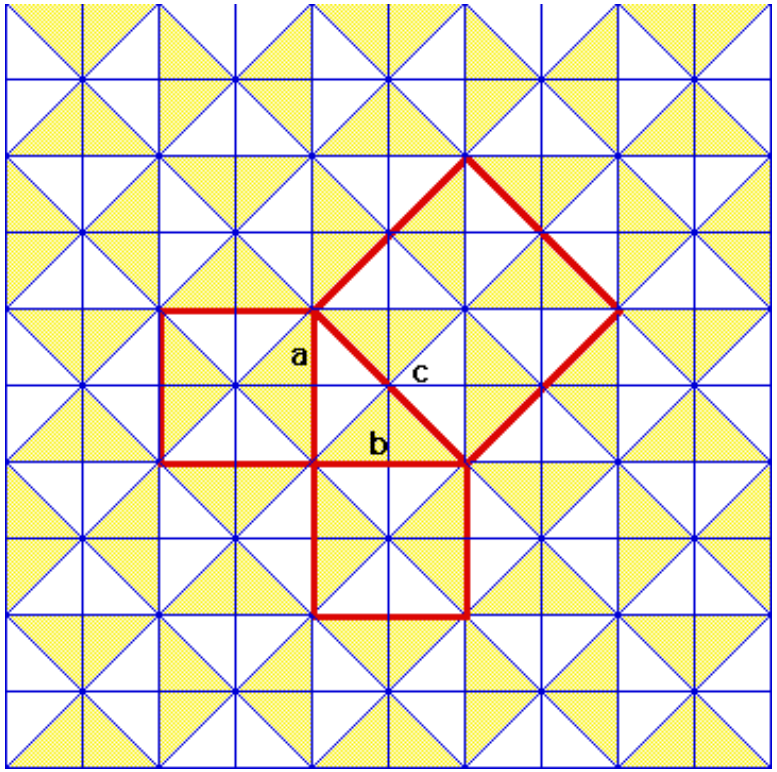
- Greek philosopher, mathematician, founder of the religious movement
- Everything is number
- Number as a fundamental aspect of reality
- Mathematics as a basic tool for investigating this reality



Pythagoras
(c. 570 – c. 495 BC)

Pythagoras' theorem

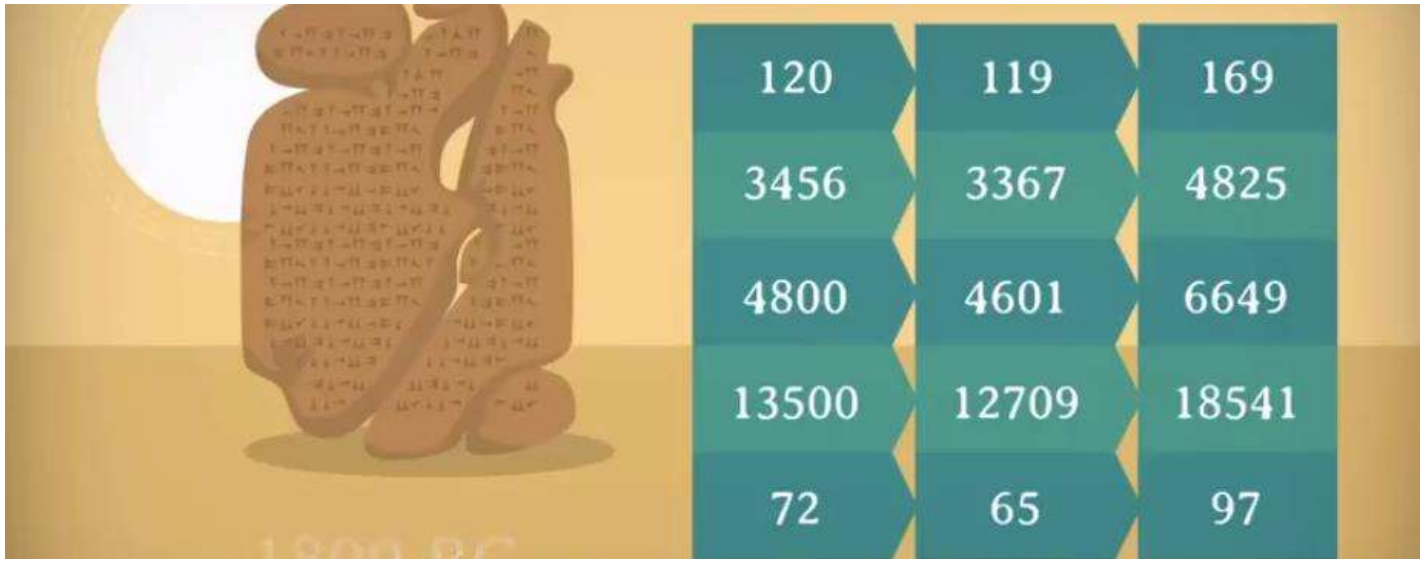
In the West, by tradition is credited with its discovery and proof to Pythagoras.



$$a^2 + b^2 = c^2$$

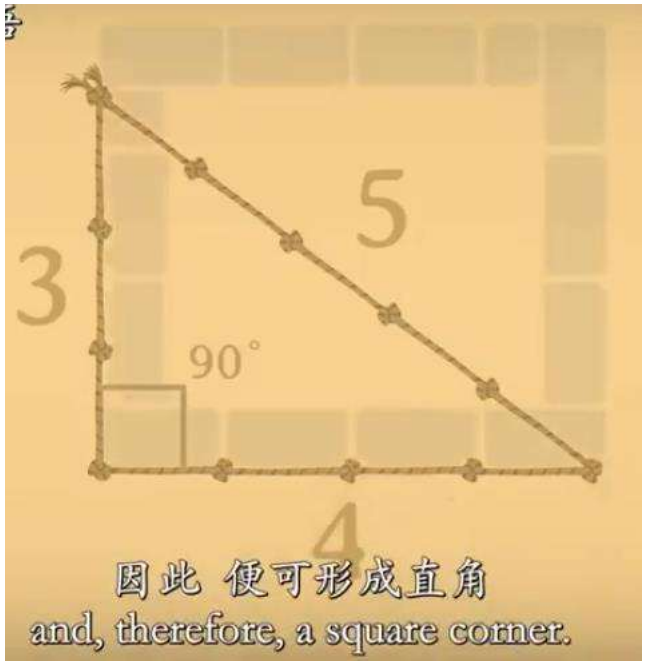
Pythagoras’ theorem

- Pythagoras (570 – 495 BC)

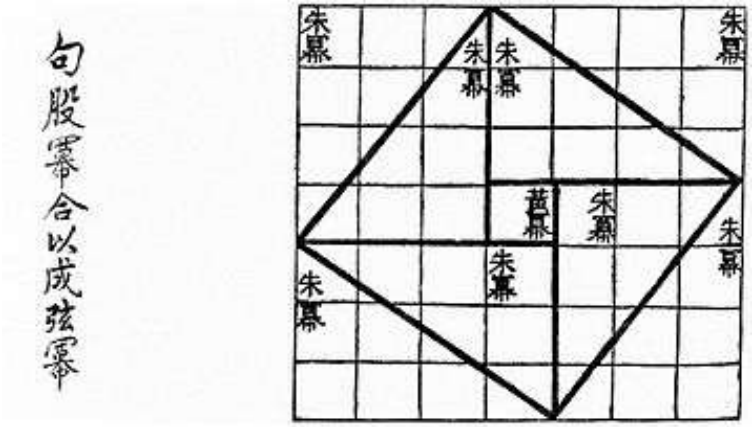


1800 BC

Babylonia tablet: 15 sets of numbers



Pythagoras' theorem



Pythagorean theorem from the 《周髀算经》

- Pythagoras' theorem <> In ancient China: 勾股定理
- 1100 BC: the Chinese developed its mathematics independently, *e.g.* the Zhou Bi Suan Jing《周髀算经》.
  - Pythagoras (570 – 495 BC)

New Promethean Revolution
(新普罗米修斯革命 430B.C.-370 B.C.)

- The First Mathematical Crisis: Incommensurability (不可测比性) (discussed in Book 10, *Elements*) *e.g.* the square root of 2 (√2).

- Three famous geometric construction (几何构建) problems:
  - Squaring the circle (圆方等积): to construct a square equal to a given circle.
  - Doubling the cube (倍立方): to construct the side of a cube whose volume is double that of a cube of given edge.
  - Trisecting an angle (三等分任意角): to trisect any angle.

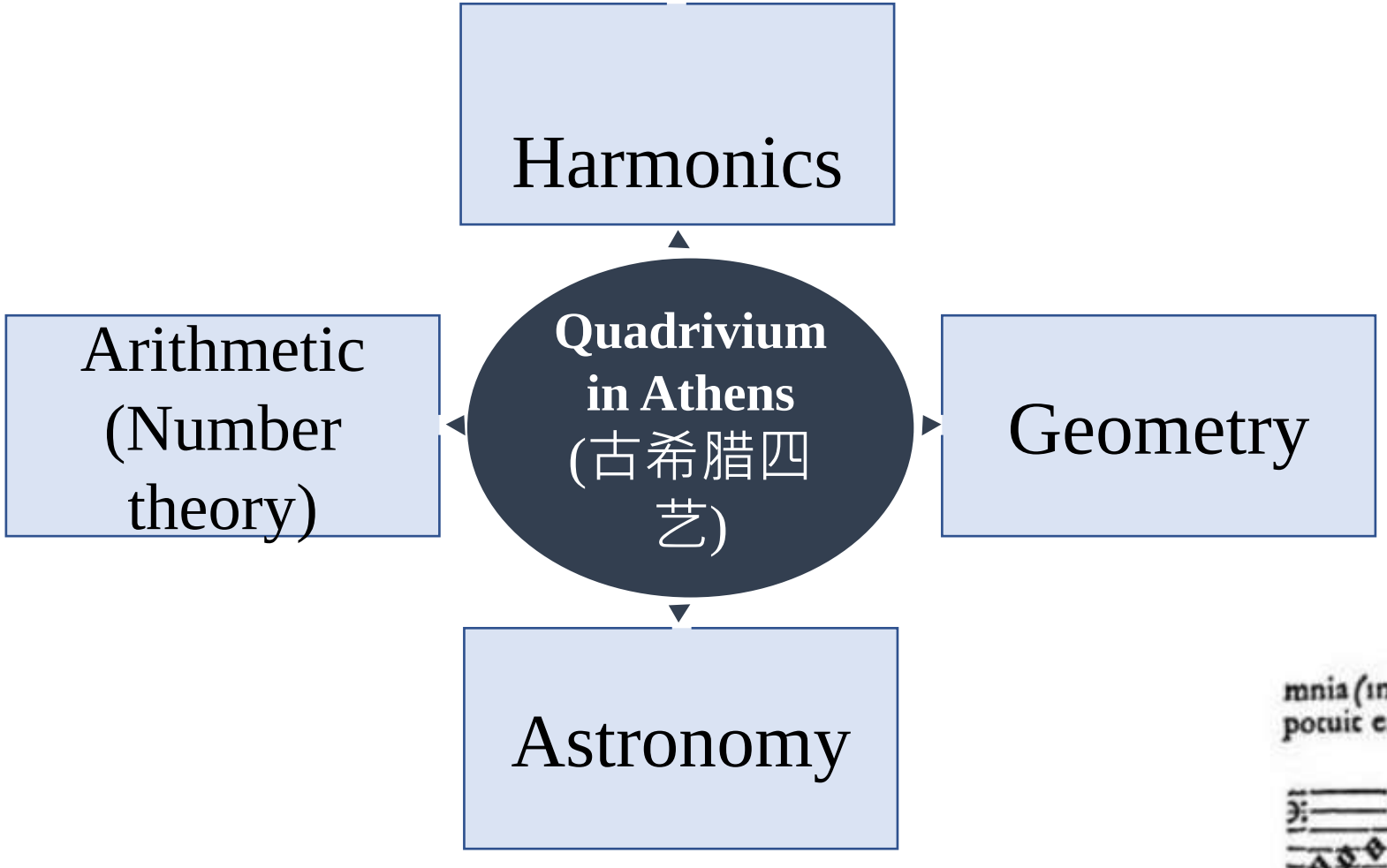
Eudoxus of Cnidus (“古希腊的牛顿”)

- Contribution of Mathematics:
  - Book 5 & 12, *Elements*
  - Theory of proportion (比例理论)
  - Method of exhaustion (穷举法)

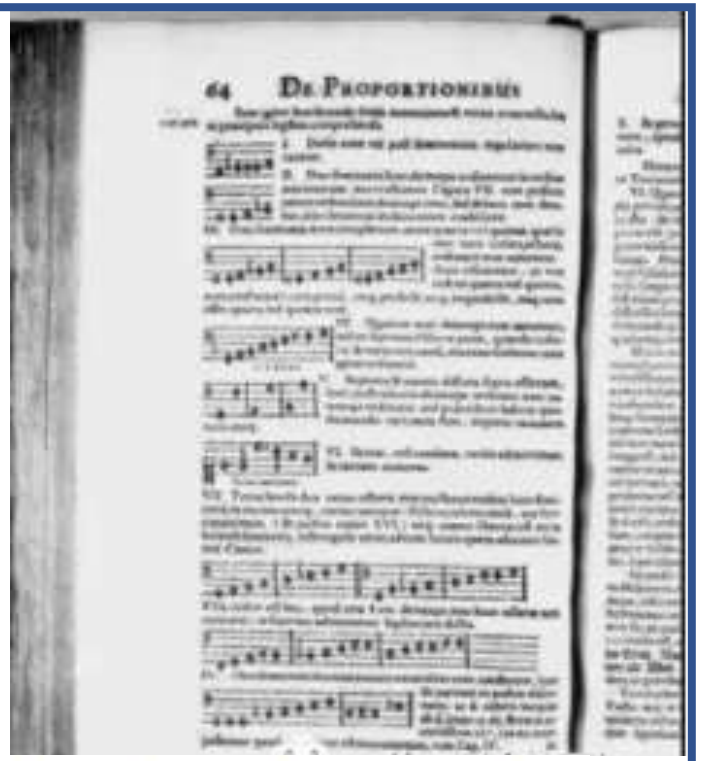
- Contribution of astronomy: (ref. para 40, chapter 3, text 2)
  - Constructed the first geometrical (几何) model of celestial motion in human history

- Laid the foundation of Greek scientific spirit-rigor of proof
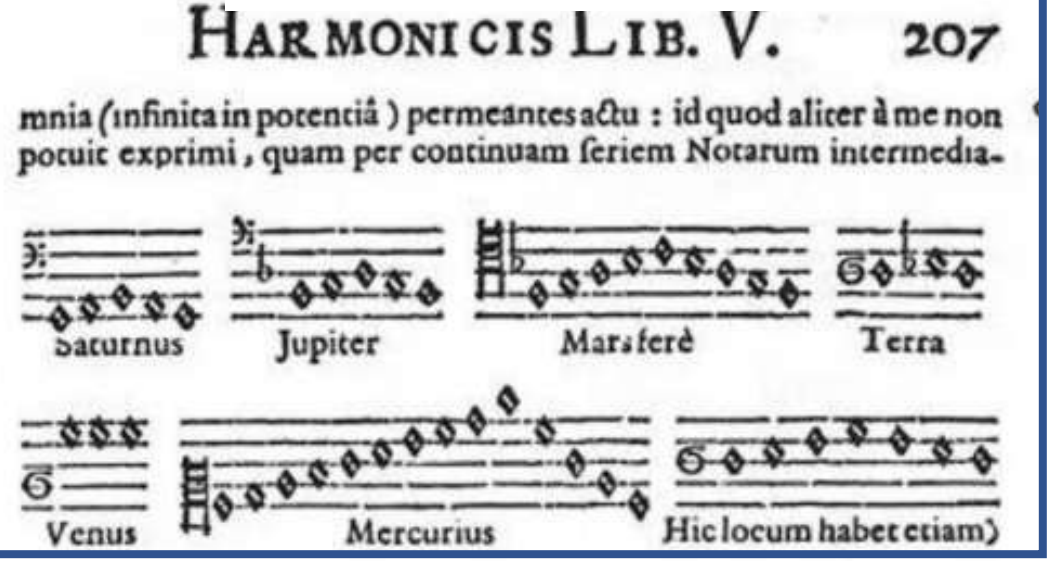
Mathematics in Ancient Greece



“Seven liberal arts (自由七艺)” in medieval universities



《世界的和谐》中用五线谱表示的行星乐音



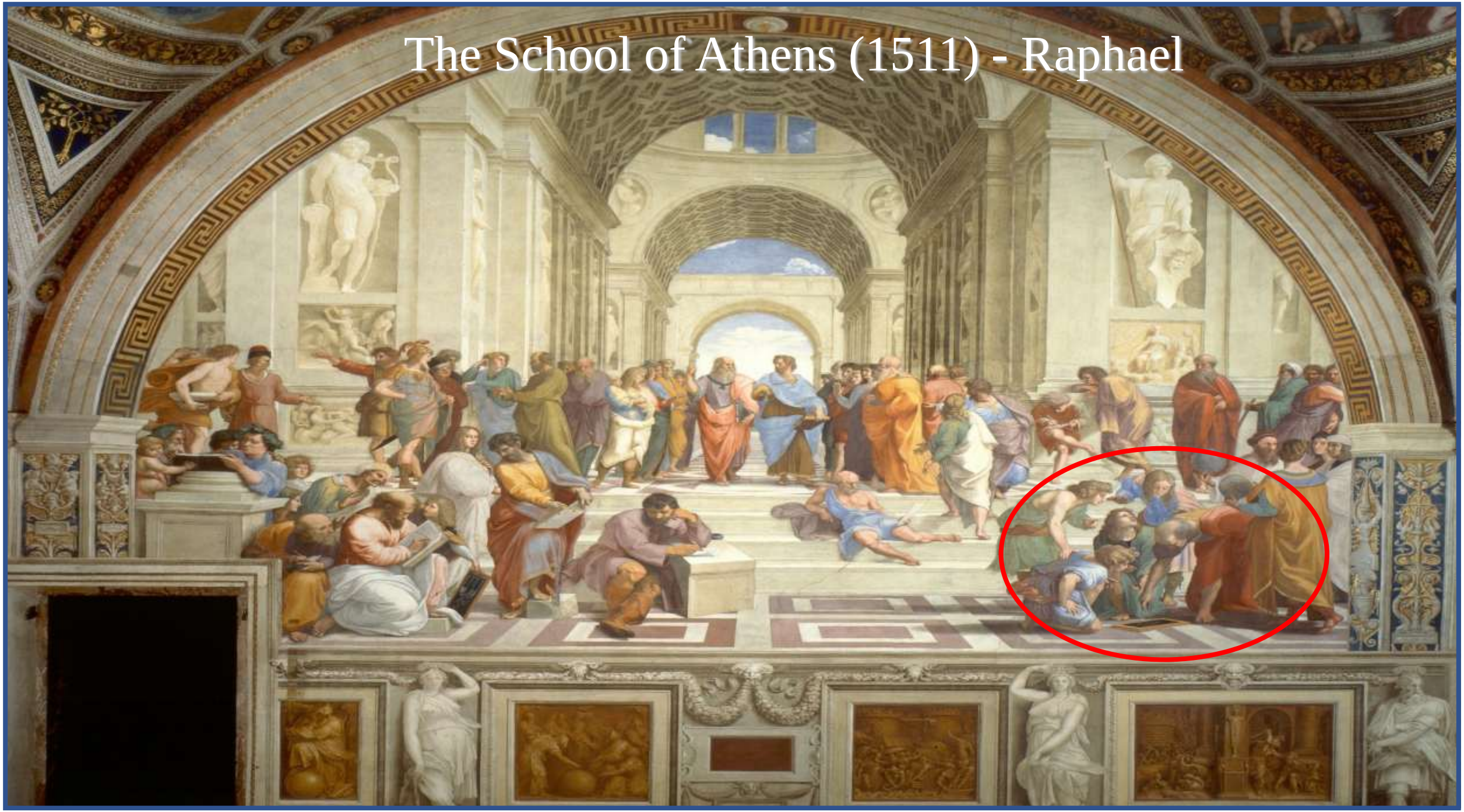
Euclid & *Elements*

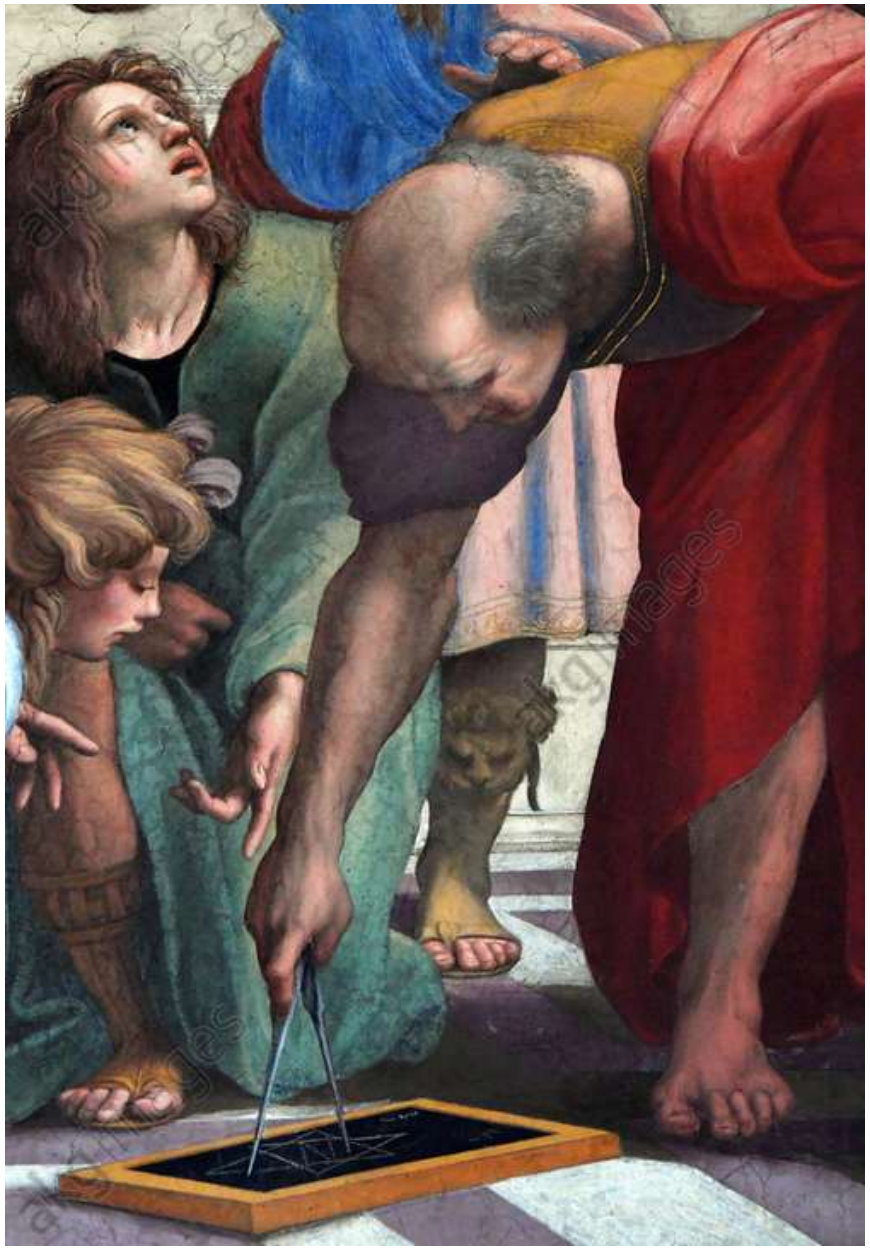
Euclid of Alexandria (欧几里得)

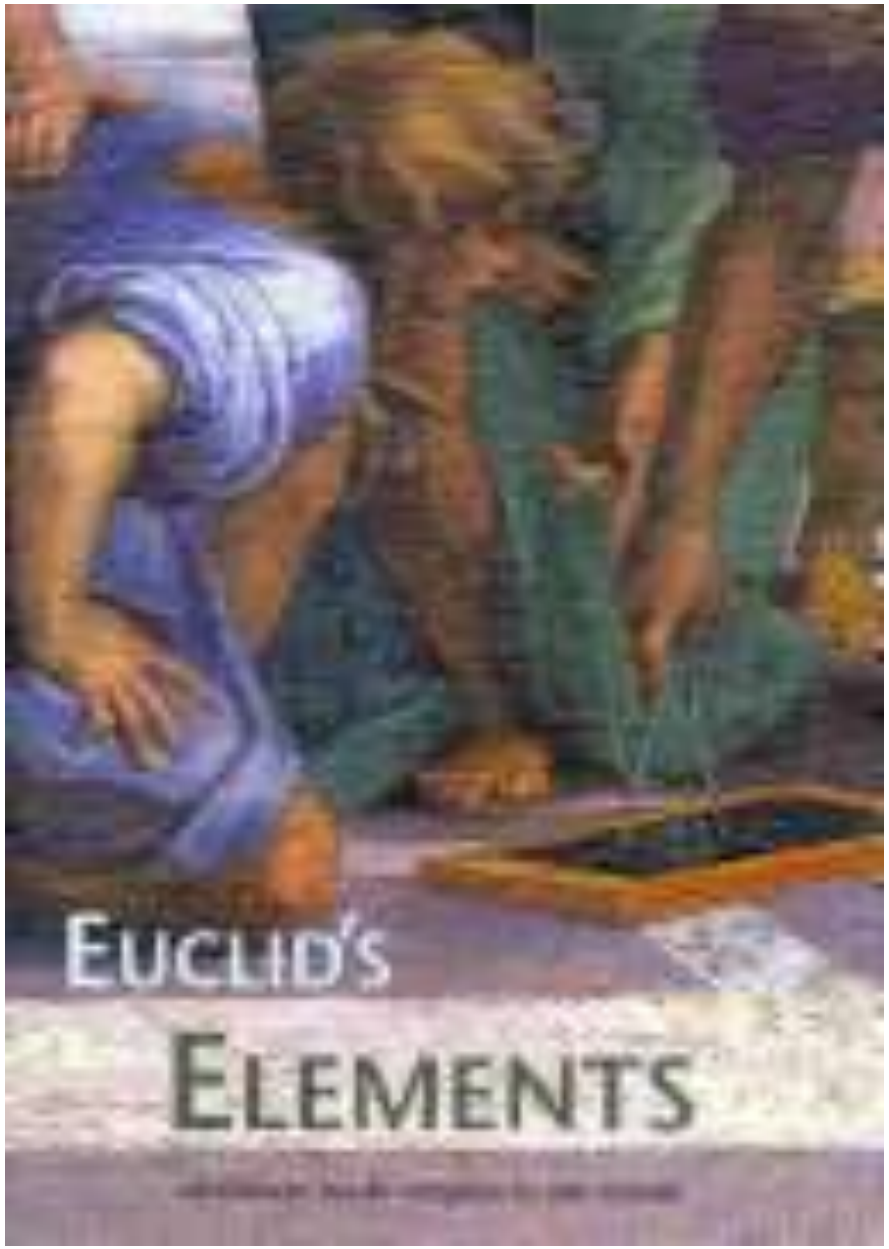


- Euclid of Alexandria (~300 BC)
  - c.f. Aristotle (384 BC – 322 BC)
  Greek mathematician, “**Father of Geometry**”.

The School of Athens (1511) - Raphael






EUCLID'S
ELEMENTS

Euclid of Alexandria (欧几里得)



- Euclid of Alexandria (~300 BC)
  - c.f. Aristotle (384 BC – 322 BC)
  Greek mathematician, "**Father of Geometry**".

- *"Euclid's Elements is called the **greatest mathematics textbook of all time**, the work has been studied ... from ancient Greece to the present day."* [Text 11a, par. 10]

- What makes the *Elements* so important is its logical development from **basic principles to sophisticated consequences**.

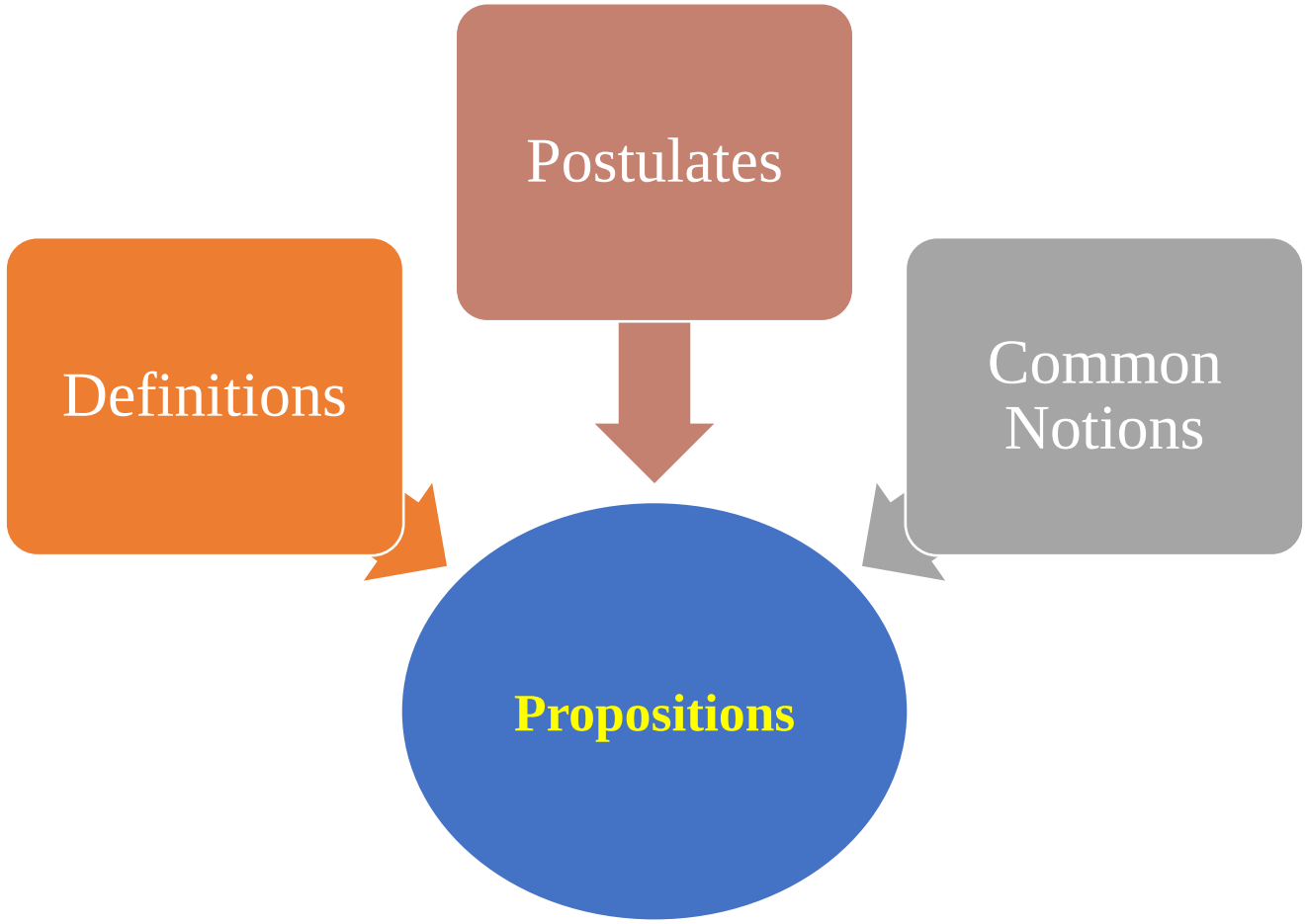
Euclid's *Elements (Green Lion Press, 2010)*

- A mathematical and geometric treatise (专著) consisting of 13 books.
- Cover plane and solid geometry, and number theory.
- Compiled (编纂) from a number of works of earlier people.
- A huge organized collection of mathematical propositions.

Euclid's *Elements*- Source & Content

Book 1	Triangles, parallels, and area	Maybe Hippocrates of Chios
Book 2	Geometric algebra	
Book 3	Circles	
Book 4	Constructions for inscribed and circumscribed figures	
Book 5	Theory of proportions	Eudoxus of Cnidus & Geometric algebra originated from ancient Babylon
Book 6	Similar figures and proportions	
Book 7	Fundamentals of number theory	Pythagorean school
Book 8	Continued proportions in number theory	
Book 9	Number theory	
Book 10	Classification of incommensurables	
Book 11	Solid geometry	Pythagorean Theaetetus & Eudoxus of Cnidus
Book 12	Measurement of figures	
Book 13	Regular solids	

Axiomatic (公理的)



Definitions (定义)

Definitions

1. A *point* is that which has no part.
2. A *line* is breadthless length.
3. The extremities of a line are points.
4. A *straight line* is a line which lies evenly with the points on itself.
5. A *surface* is that which has length and breadth only.
6. The extremities of a surface are lines.
7. A *plane surface* is a surface which lies evenly with the straight lines on itself.
8. A *plane angle* is the inclination to one another of two lines in a plane which meet one another and do not lie in a straight line.
9. And when the lines containing the angle are straight, the angle is called *rectilineal*.
10. When a straight line set up on a straight line makes the adjacent angles equal to one another, each of the equal angles is *right*, and the straight line standing on the other is called a *perpendicular* to that on which it stands.
11. An *obtuse angle* is an angle greater than a right angle.
12. An *acute angle* is an angle less than a right angle.
13. A *boundary* is that which is an extremity of anything.
14. A *figure* is that which is contained by any boundary or boundaries.
15. A *circle* is a plane figure contained by one line such that all the straight lines falling upon it from one point among those lying within the figure are equal to one another;
16. And the point is called the *centre* of the circle.
17. A *diameter* of the circle is any straight line drawn through the centre and terminated in both directions by the circumference of the circle, and such a straight line also bisects the circle.
18. A *semicircle* is the figure contained by the diameter and the circumference cut off by it. And the centre of the semicircle is the same as that of the circle.

- Euclid began Book I with a list of **23 definitions**.
- So the reader would know precisely what his terms meant – leaving **no room for ambiguity (模棱两可)**.

Postulates (公设) and Common Notions (公理)

Postulates

Let the following be postulated:

1. To draw a straight line from any point to any point.
2. To produce a finite straight line continuously in a straight line.
3. To describe a circle with any centre and distance.
4. That all right angles are equal to one another.
5. That, if a straight line falling on two straight lines make the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which are the angles less than the two right angles.

Common Notions

1. Things which are equal to the same thing are also equal to one another.
2. If equals be added to equals, the wholes are equal.
3. If equals be subtracted from equals, the remainders are equal.
4. Things which coincide with one another are equal to one another.
5. The whole is greater than the part.

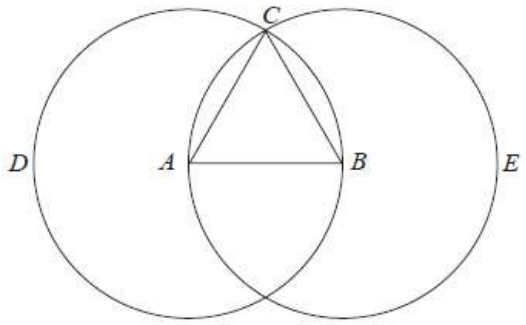
- **Postulates (or axioms) or self-evident truth:**
  Only 5
- **Common Notions:**
  Self-evident statements, i.e. common senses
- ***"These were given without proof or justification. They were simply to be accepted."***
- **These are beliefs, but are they convincing?**

Propositions (命題)

Propositions

Proposition 1.

On a given finite straight line to construct an equilateral triangle.



Let *AB* be the given finite straight line.

Thus it is required to construct an equilateral triangle on the straight line *AB*.

With centre *A* and distance *AB* let the circle *BCD* be described; [Post. 3]

again, with centre *B* and distance *BA* let the circle *ACE* be described; [Post. 3]

and from the point *C*, in which the circles cut one another, to the points *A*, *B* let the straight lines *CA*, *CB* be joined. [Post. 1]

Now, since the point *A* is the centre of the circle *CDB*,

AC is equal to *AB*. [Def. 15]

Again, since the point *B* is the centre of the circle *CAE*,

BC is equal to *BA*. [Def. 15]

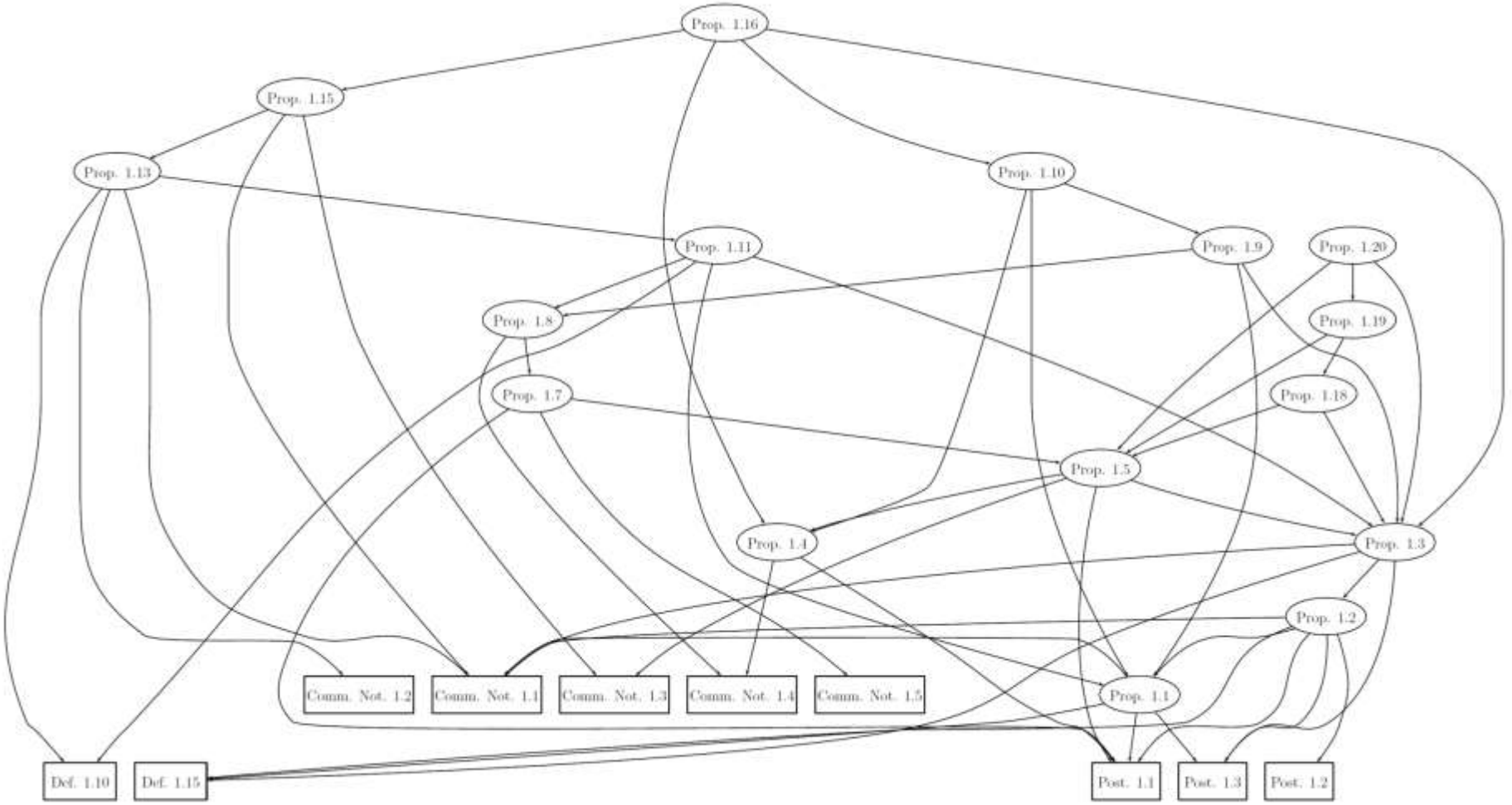
But *CA* was also proved equal to *AB*;

therefore each of the straight lines *CA*, *CB* is equal to *AB*.

Propositions:

- Deduced from definitions, postulates, common notions, and other propositions.
- Compare to the postulates or common notions, propositions are sometimes **not straight forward** or even **violates our common senses**.

Axiomatic System



The definitions, postulates and propositions are closely related.

“Non-Euclidean” geometry (非欧几何, 19th century)

flat curvature **positive curvature** **negative curvature**

=180 **>180** **<180**



- How did you learn geometry in high school?
- How did Euclid do geometry with a collapsing compass and straightedge (ruler) only?





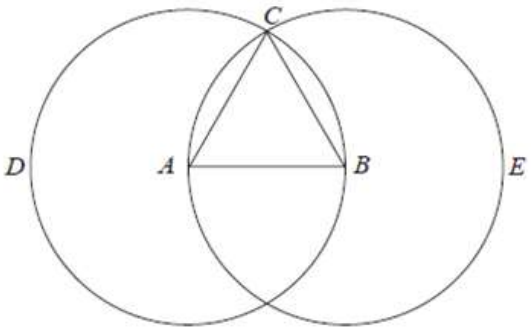
1. Construct an equilateral triangle with a collapsible compass (Postulate 1)



Propositions

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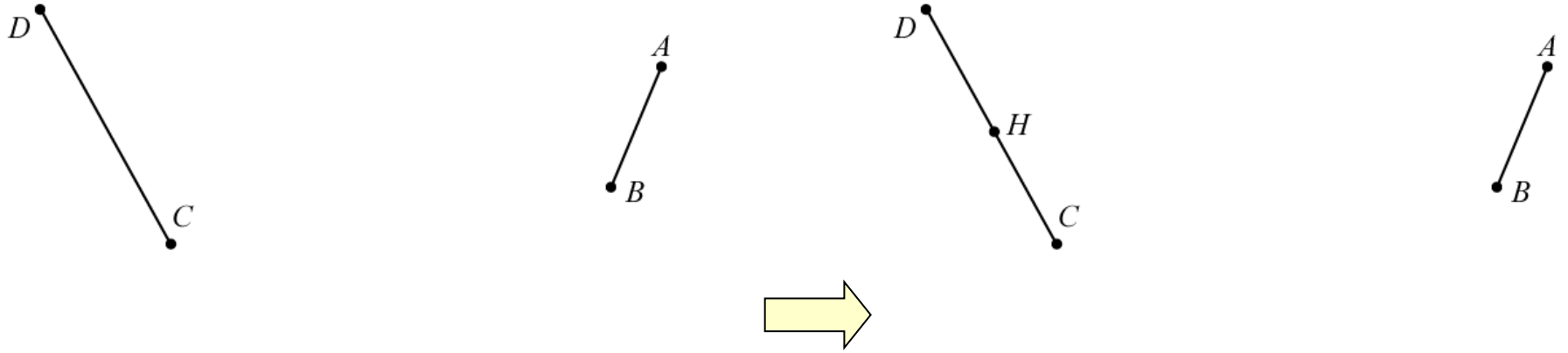
Again, since the point *B* is the centre of the circle *CAE*,

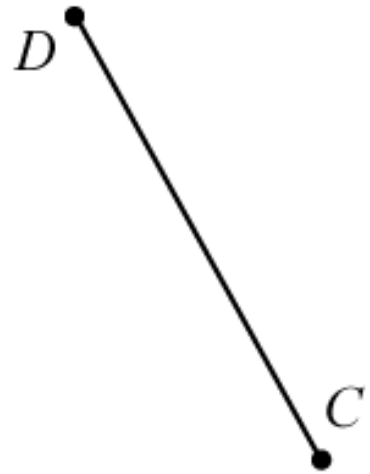
BC is equal to *BA*. [Def. 15]

But *CA* was also proved equal to *AB*;

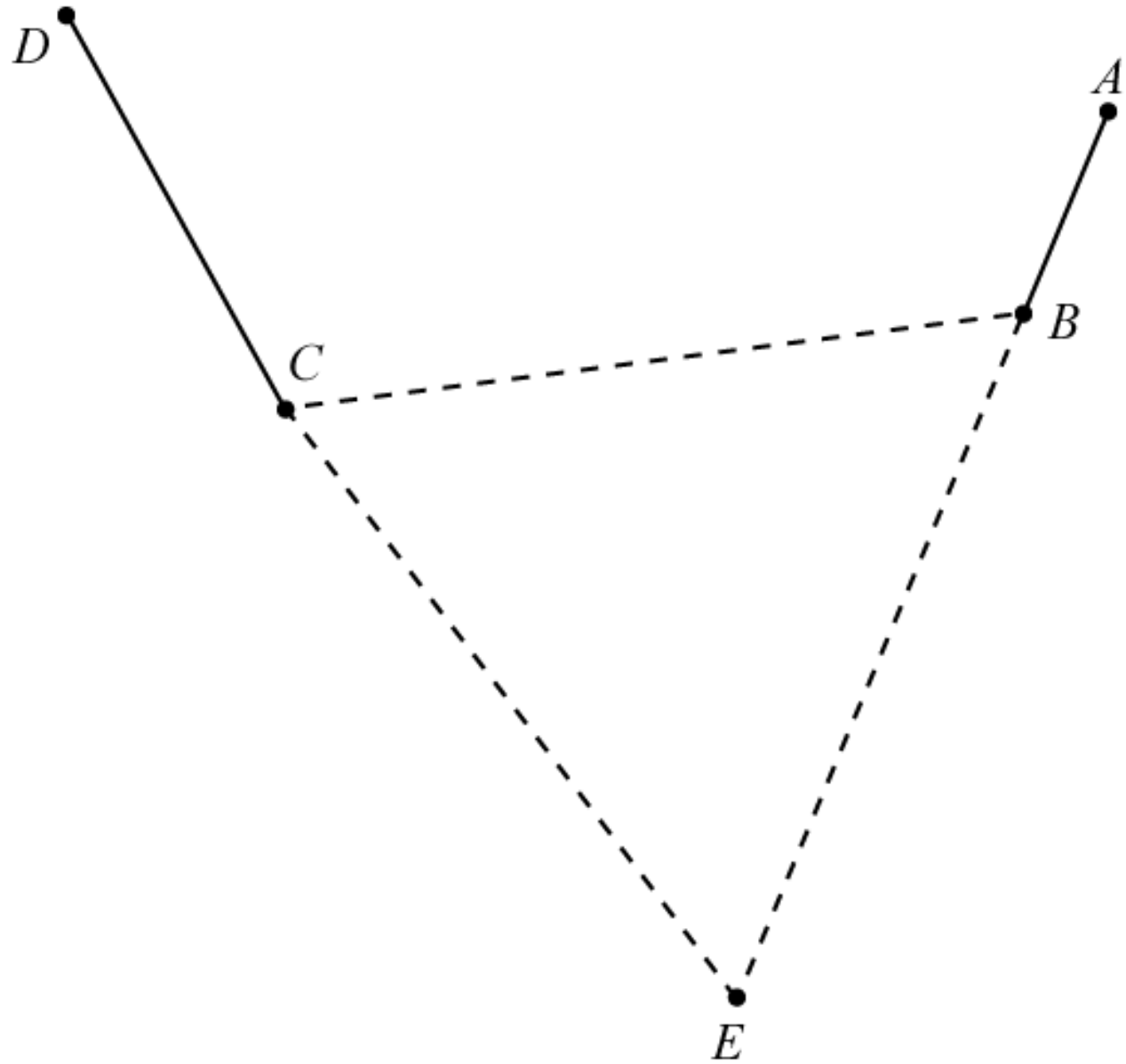
therefore each of the straight lines *CA*, *CB* is equal to *AB*.

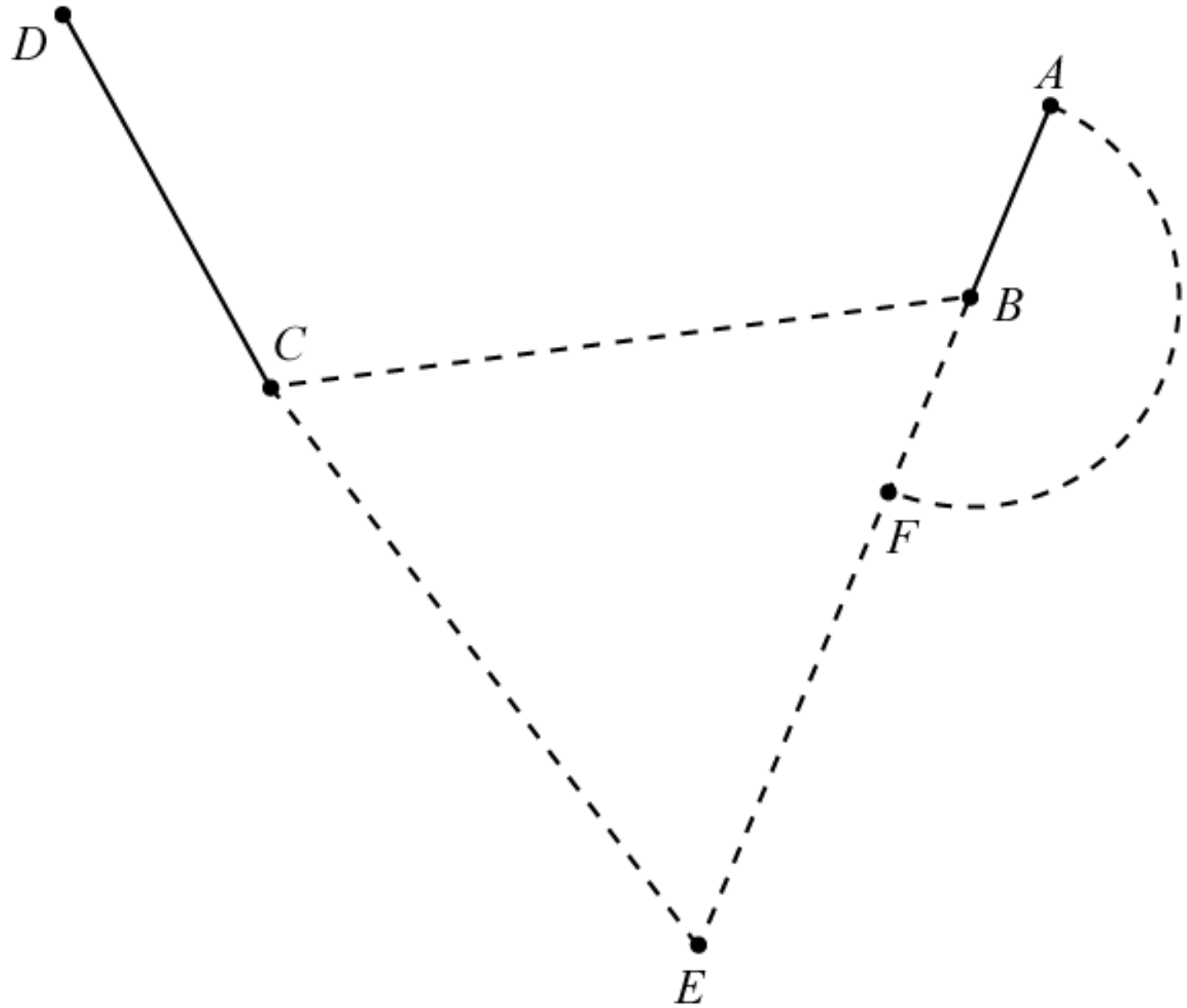
2. Transfer a length to another line

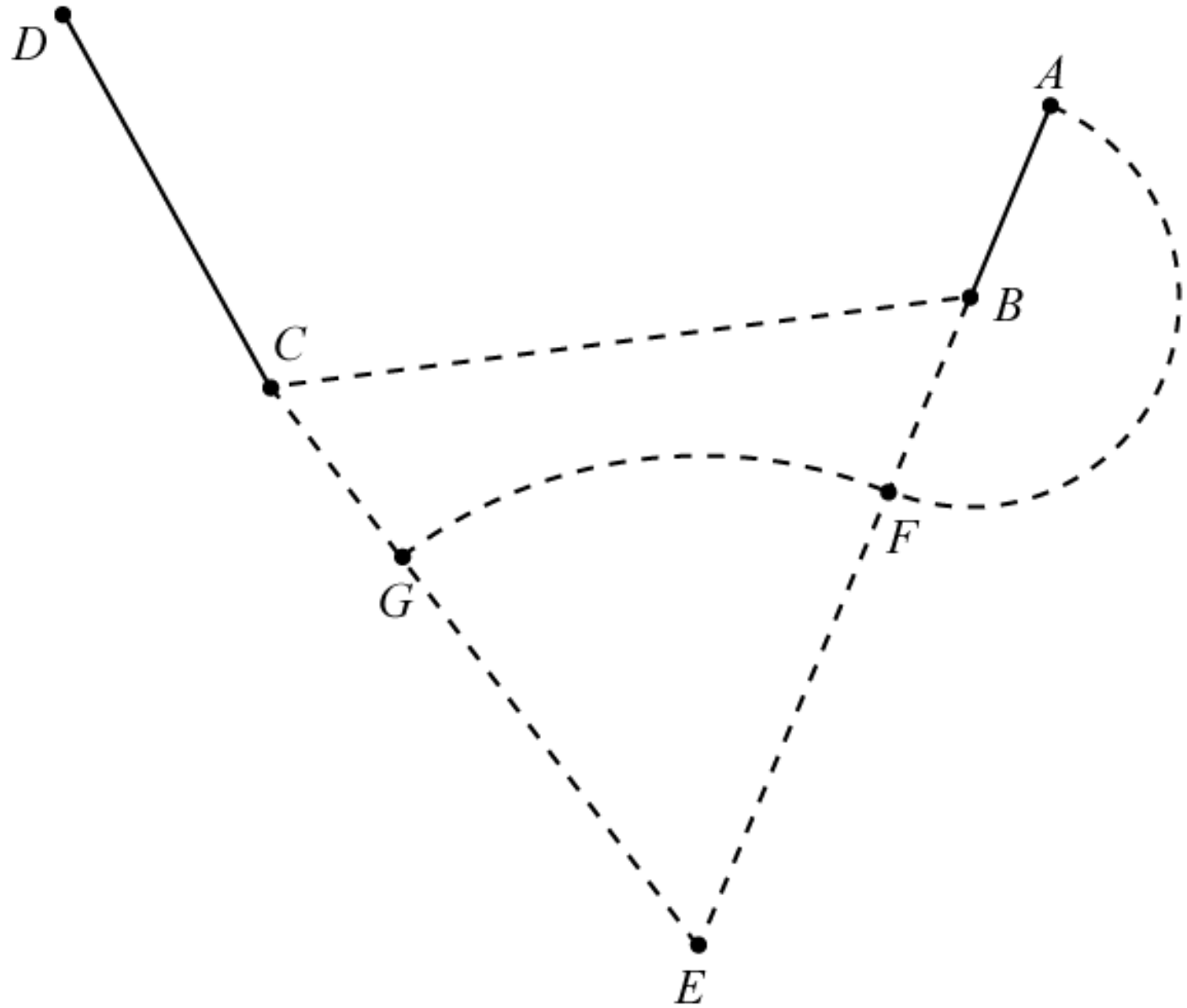


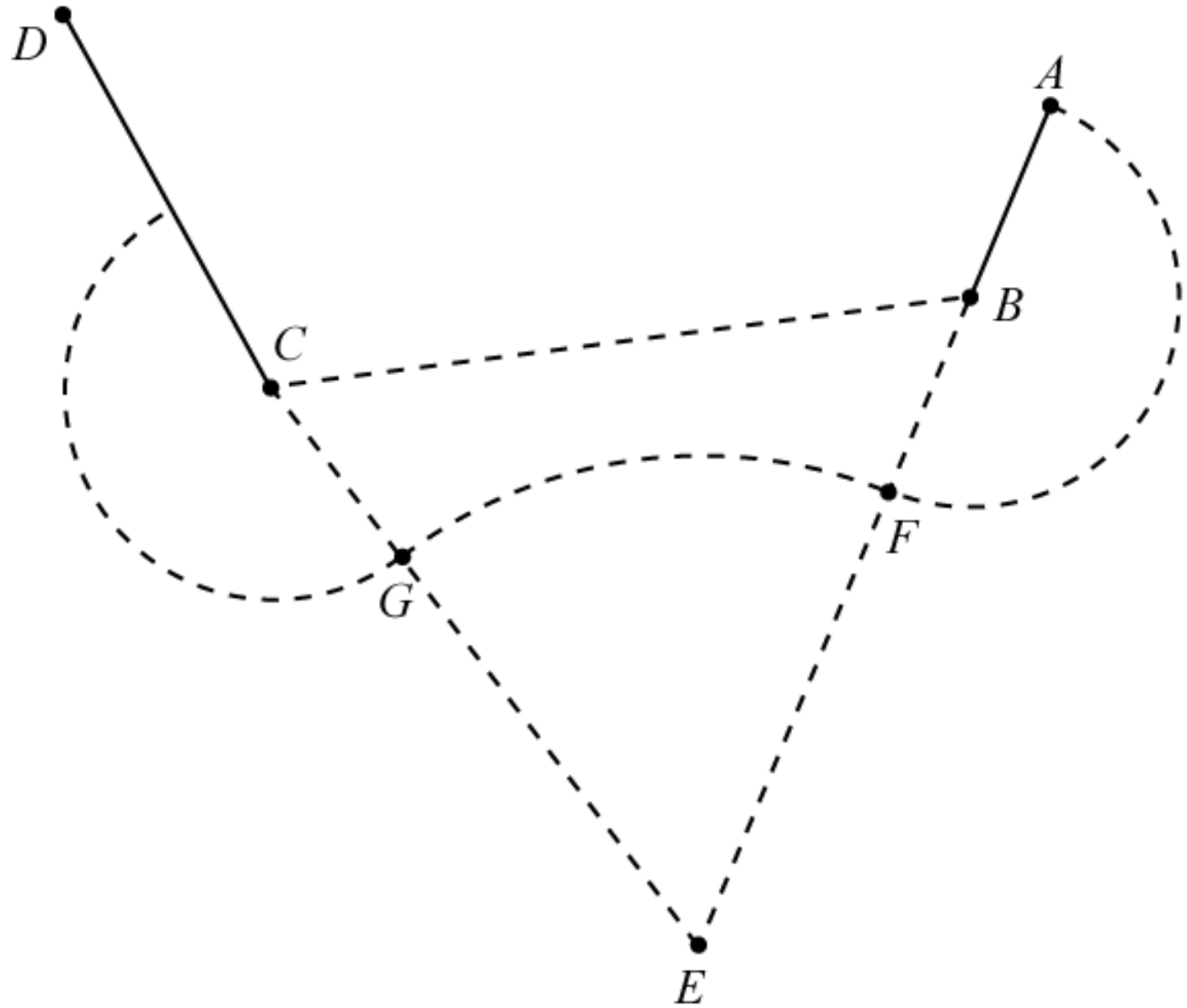

D
C

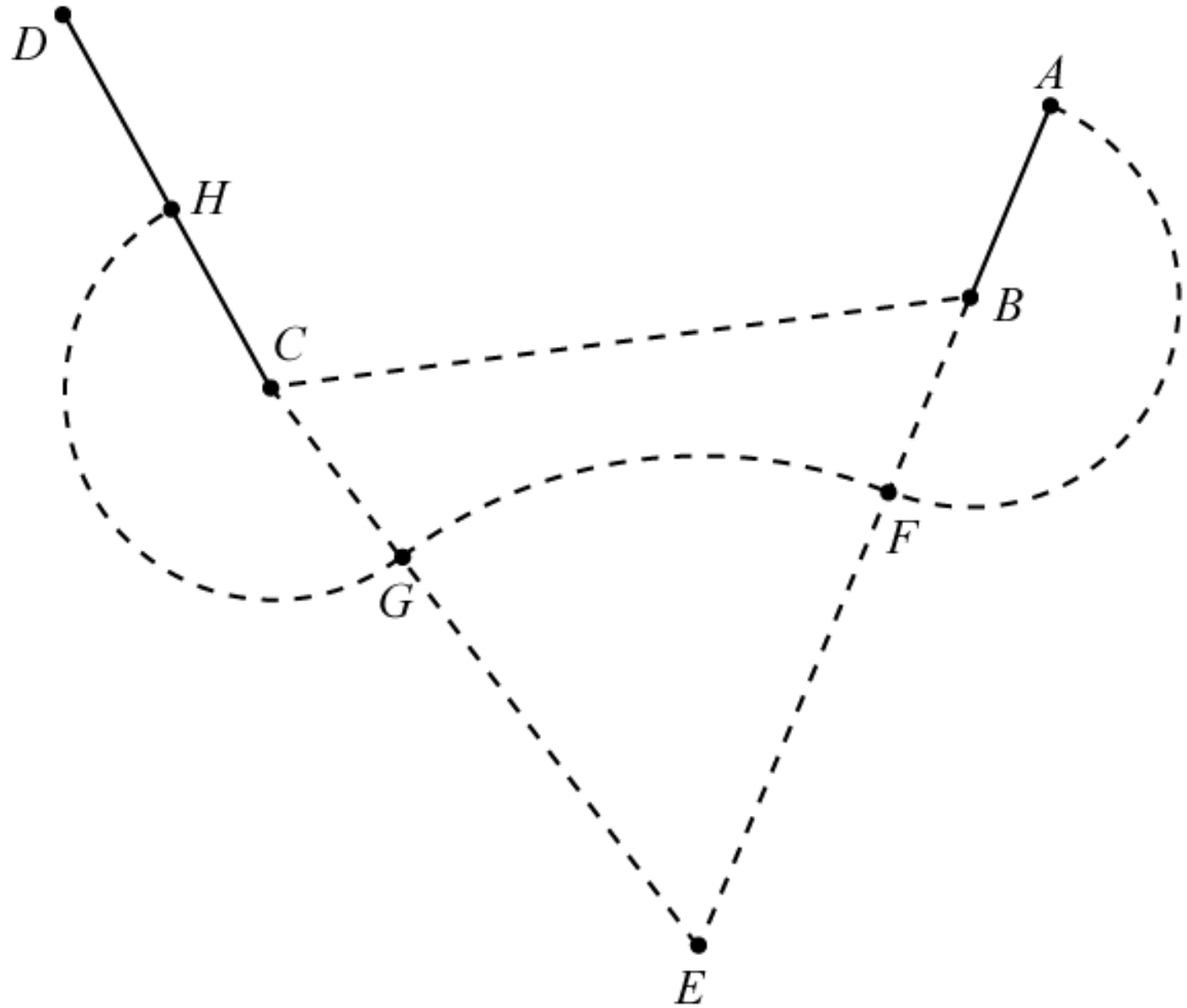

A
B

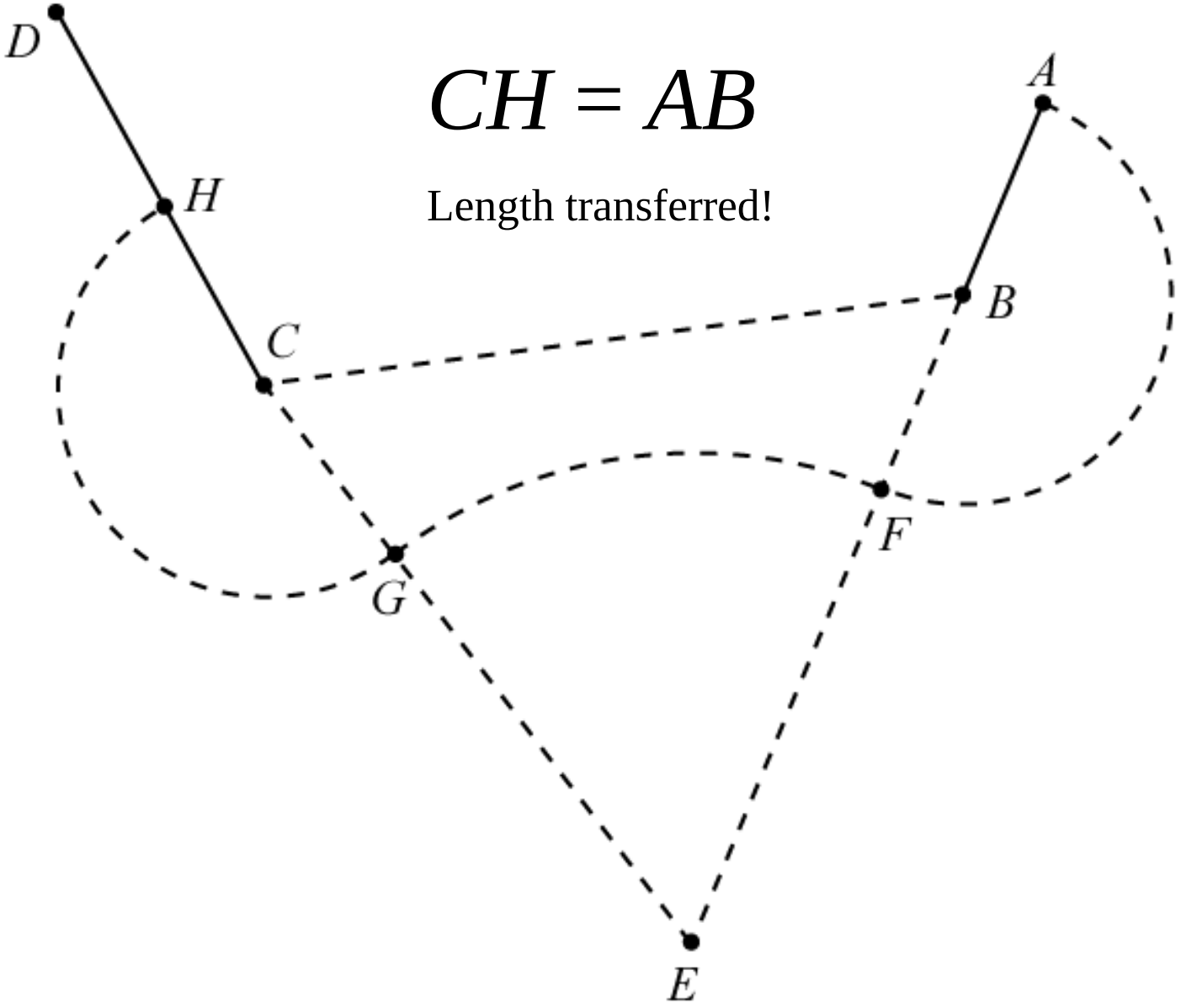

D
A
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E


D
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D
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D
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D
H
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A
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G
E



$$CH = AB$$

Length transferred!

Characteristics of Greek Geometry

- Reliance on collapsing compass and straightedge (ruler) only
- Impressive logical rigor
- Purely geometric, as opposed to numerical, nature of their mathematics
- Skillful organization
- Pure thought, ideal, immaterial and eternal

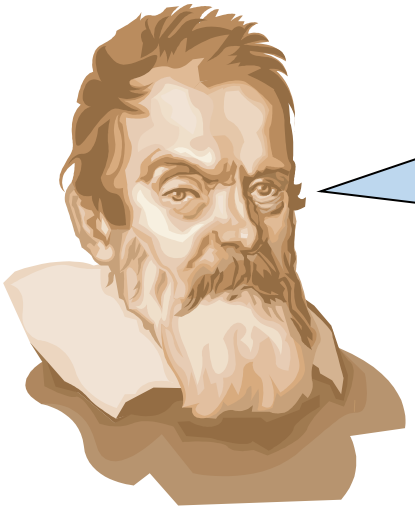




Elements & Modern Science

Science and Mathematics

- What does mathematics to do with the physical world?
- There is deep connection between science and mathematics.



Galileo Galilei

The universe cannot be read until we have learned the language and become familiar with the characters in which it is written. It is **written in mathematical language**, and the letters are triangles, circles and other geometrical figures.

The miracle of mathematics in the natural sciences

- **Eugene Wigner (维格纳),** an American Nobel laureate in Physics, wrote *The Unreasonable Effectiveness of Mathematics in the Natural Sciences* in 1960.



Eugene Wigner
(1902-1995)

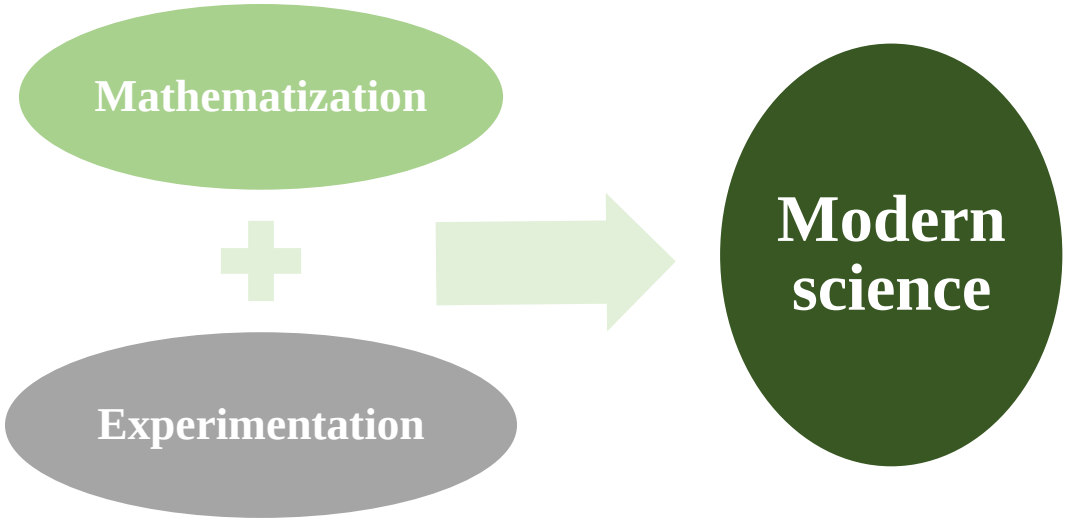
- Is this a miracle? How can **mathematics**, something that "seems to borrow least from the exterior world", **explain the nature so perfectly**?
- He argued that this miracle, the laws of nature, is highly related to the **miracle of the human mind**. These two miracles are probably the same miracle!

Needham Question

Why did modern science emerge in Western Europe rather than in China, despite the earlier predominance of Chinese science and technology?

- Needham's definition of "modern science"

"If I were asked to define modern science, I would say that it was the combination of ***mathematized*** *hypotheses about natural phenomena with relentless* ***experimentation****"*



Lack of Geometrical Proof

- "Let us first list some of the handicaps to the development of modern science which are found in the early and medieval civilization of China. Of the greatest importance of all, perhaps, is the fact that the Chinese did not develop as far as the Greeks the idea of **geometrical proof**."

(SCC7:2 p. 210)

"One may also speculate on whether Western Europe would have elaborated scientific method without **Euclid**…" (SCC7:2 p. 231)

SCC7:2 – *Science and Civilisation in China*, Vol. 7, Part 2 (Needham)

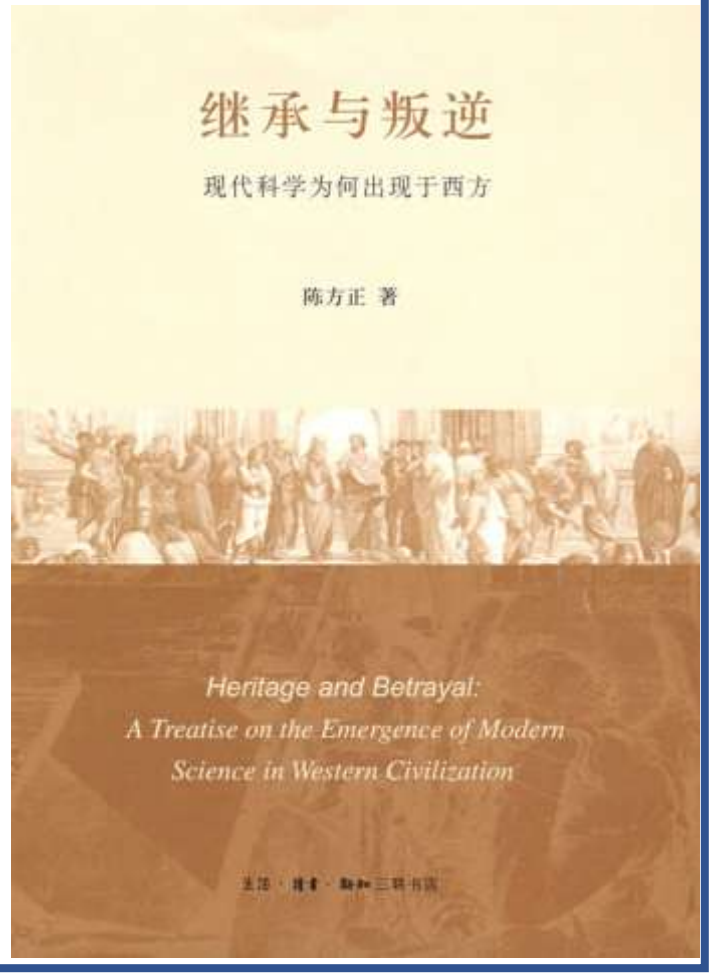
Heritage and Betrayal



- 陈方正：《继承与叛逆：现代科学为何出现于西方》(2009)

  Heritage and Betrayal: A Treatise on the Emergence of Modern Science in Western Civilization

- Chen asked, “Why did modern science emerge in Western civilization?”



- Chen argued that ancient Chinese science was not as advanced as Needham believed. Therefore, Needham's question is out of question.
- Factors that led to the emergence of modern science in Western civilization were
  - A long tradition of heritage and betrayal 继承与叛逆
    (c.f. Texts 1, 2 and 3)
  - The beginnings of **mathematics** in Egypt and ancient Babylonia
    - Started 1500 years earlier than in ancient China.
  - The two revolutions that led to ancient Greek science and Newtonian physics.
    - Euclid's *Elements* (Text 11b) + Newton's *Principia* (Text 3b)

Influence on Newton

Text 11b

from

Elements

by Euclid

BOOK I

Definitions

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11. An *obtuse angle* is an angle greater than a right angle.
12. An *acute angle* is an angle less than a right angle.

Text 3b

from

The Principia: Mathematical Principles of Natural Philosophy

by Isaac Newton

DEFINITIONS.

Definition 1

Quantity of matter is a measure of matter that arises from its density and volume jointly.

If the density of air is doubled in a space that is also doubled, there is four times as much air, and there is six times as much if the space is tripled. The case is the same for snow and powders condensed by compression or liquefaction, and also for all bodies that are condensed in various ways by any causes whatsoever. For the present, I am not taking into account any medium, if there should be any, freely pervading the interstices between the parts of bodies. Furthermore, I mean this quantity whenever I use the term "body" or "mass" in the following pages. It can always be known from a body's weight, for—by making very accurate experiments with pendulums—I have found it to be proportional to the weight, as will be shown below.

Euclid

Postulates

Let the following be postulated:

1. To draw a straight line from any point to any point.
2. To produce a finite straight line continuously in a straight line.
3. To describe a circle with any centre and distance.
4. That all right angles are equal to one another.
5. That, if a straight line falling on two straight lines make the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which are the angles less than the two right angles.

Common Notions

1. Things which are equal to the same thing are also equal to one another.
2. If equals be added to equals, the wholes are equal.
3. If equals be subtracted from equals, the remainders are equal.
4. Things which coincide with one another are equal to one another.
5. The whole is greater than the part.

Newton

AXIOMS, OR THE LAWS OF MOTION

Law 1

Every body perseveres in its state of being at rest or of moving uniformly straight forward, except insofar as it is compelled to change its state by forces impressed.

Projectiles persevere in their motions, except insofar as they are retarded by the resistance of the air and are impelled downward by the force of gravity. A spinning hoop, which has parts that by their cohesion continually draw one another back from rectilinear motions, does not cease to rotate, except insofar as it is retarded by the air. And larger bodies—planets and comets—preserve for a longer time both their progressive and their circular motions, which take place in spaces having less resistance.

Law 2

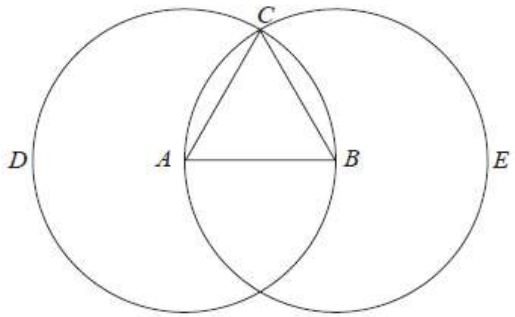
A change in motion is proportional to the motive force impressed and takes place along the straight line in which that force is impressed.

Euclid

Propositions

Proposition 1.

On a given finite straight line to construct an equilateral triangle.



Let *AB* be the given finite straight line.

Thus it is required to construct an equilateral triangle on the straight line *AB*.

With centre *A* and distance *AB* let the circle *BCD* be described; [Post. 3]

again, with centre *B* and distance *BA* let the circle *ACE* be described; [Post. 3]

and from the point *C*, in which the circles cut one another, to the points *A*, *B* let the straight lines *CA*, *CB* be joined. [Post. 1]

Now, since the point *A* is the centre of the circle *CDB*,

AC is equal to *AB*. [Def. 15]

Again, since the point *B* is the centre of the circle *CAE*,

BC is equal to *BA*. [Def. 15]

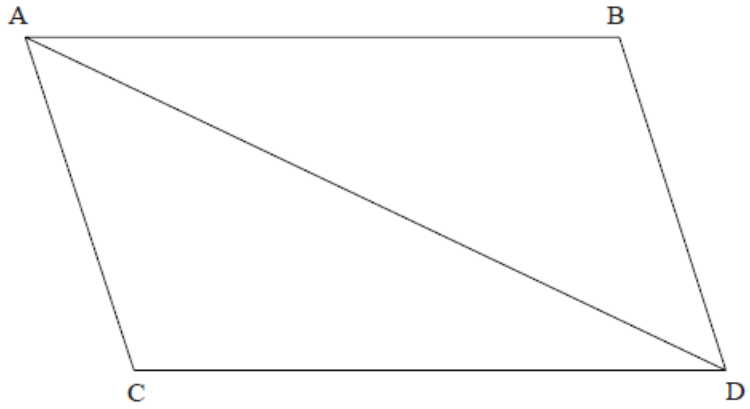
But *CA* was also proved equal to *AB*;

therefore each of the straight lines *CA*, *CB* is equal to *AB*.

Newton

Corollary 1

A body acted on by [two] forces acting jointly describes the diagonal of a parallelogram in the same time in which it would describe the sides if the forces were acting separately.



Let a body in a given time, by force M alone impressed in A, be carried with uniform motion from A to B, and, by force N alone impressed in the same place, be carried from A to C; then complete the parallelogram ABDC, and by

Comparison

Axiomatic systems

Elements

- Definitions
- Postulates

⇨ A system of mathematics

Principia

- Definitions
- Axioms

⇨ A system of mechanics

Mathematical and Axiomatic

Euclid's influence on Science

- *"It is the glory of geometry that so much is accomplished with so few principles that are obtained elsewhere."*
  - **Isaac Newton**, Preface of *Principia*

- *"If Euclid failed to kindle your youthful enthusiasm, then you were not born to be a scientific thinker."*
  - **Albert Einstein**, Text 11a, para. 45

- *"I was taught Euclid by a private tutor, and I distinctly remember the intense satisfaction which the clear geometrical proofs gave me."*
  - **Charles Darwin**, *Autobiography Of Charles Darwin*

Translated by Xu Guangqi (徐光启)



➢Translated the *Elements* into Chinese with Matteo Ricci (利玛窦) in late Ming Dynasty.

徐光启评《几何原本》

- 此书有四不必:
  - 不必疑，不必揣，
  - 不必试，不必改。
- 有四不可得:
  - 欲脱之不可得，欲驳之不可得，
  - 欲减之不可得，欲前后更置之不可得。
- 有三至三能:
  - 似至晦，实至明，故能以其明明他物之至晦；
  - 似至繁，实至简，故能以其简简他物之至繁；
  - 似至难，实至易，故能以其易易他物之至难。
- 易生于简，简生于明，综其妙在明而矣。



Part III Our Understanding of Human Understanding

What is science? Knowledge about science

- Nature of science
  - Testability/falsifiability
  - Causal explanation (mechanical cause and effect)
  - Objectivity, limitation *e.g.* subjectivity
- Scientific method
  - Reductionism
  - Experimentation
  - Mathematization
- Science and culture
  - Modern science vs. Chinese science

Why science?
Values of science

- Intellectual beauty
- Enduring question
  - What is mind?
  - Who am I?
  - Do we have free will?

TP consultation

- Optional
- Place: TB516
- https://docs.qq.com/sheet/DTk9FWGZVZndBS2hM?tab=BB08J2&_t=1744789818253&nlc=1&u=a0b3f207362a4077ae8007345e1f4ba8
- Please come with your outline.

Reading Assignment



Finish reading Text 11.